PRAJAKTA SHEVAKARI

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PROFESSIONAL SUMMARY

- Master's candidate in CS specializing in AI/ML and NLP, with six years of experience in backend and distributed systems.
- Designed and implemented end-to-end machine learning pipelines through academic and applied projects, spanning data ingestion, model training, evaluation, and system integration, with measurable results on real-world datasets.

EDUCATION

Johns Hopkins University — ***August 2024 - May 2026***

Master's, Computer Science — *GPA: 3.7*

- Artificial Intelligence, Machine Learning, Natural Language Processing, ML: AI System Design & Development, Human Language Technology, Software System Design, Parallel Computing for Data Science

Savitribai Phule Pune University — *August 2013 - May 2017*

Bachelor's, Computer Science — *GPA: 3.8*

- Data Structures and Problem Solving, Cloud Computing, Business Analytics and Intelligence, High-Performance Computing, Computer Networks, Advanced Computer Programming, Design and Analysis of Algorithms

MACHINE LEARNING PROJECTS

Machine Learning Engineer, Johns Hopkins University — *Aug 2024 - Dec 2025*

D-FuseSmileNet - Deep Learning for Spontaneous Smile Recognition:

- Built a PyTorch pipeline combining transformer features and physiologically grounded embeddings for spontaneous vs posed smiles.
- Evaluated cross-dataset generalization on UvA-NEMO, MMI, SPOS, and BBC (1,240+ videos).
- Achieved 96.73% average accuracy (88.7%–100%), demonstrating robustness across datasets.

Clinical NER with LLMs and CoT Prompting:

- Built an automated annotation pipeline using GPT-4-mini with chain-of-thought prompting and parameter-efficient fine-tuning.
- Generated synthetic labels for low resource clinical data (31,783 samples across 7 entity types), reducing manual labeling.
- Improved token-level F1 from 0.120 to 0.335; fine-tuned BanglishBERT achieved macro-F1 of 46.0.

TerraNova – AI-Powered Post-Wildfire Decision Support System:

- Built a three-stage ML system with U-Net damage segmentation, Random Forest reburn-risk modeling, and GA-based planning.
- Trained models on thousands of multispectral image patches and integrated risk prediction with multi-objective optimization.
- Achieved ROC-AUC 0.9912, F1 0.9888, and accuracy 0.9867; optimization improved intervention scores by +0.025 in 3.46s.

TECHNICAL SKILLS

- **Machine Learning & Modeling:** PyTorch, TensorFlow, Scikit-learn, Computer Vision (CNNs, U-Net), Model Evaluation (F1, ROC-AUC)
- **NLP & LLMs:** Hugging Face Transformers, LLM fine-tuning, Chain-of-Thought Prompting, Synthetic Data Generation
- **MLOps & Data Engineering:** Docker, Kubernetes, Apache Spark, ETL, ML Training/Serving/Monitoring
- **Programming:** Python, Java, Bash, SQL
- **Cloud & Systems:** AWS (EC2, S3, Lambda), Microservices, REST APIs, System Design
- **Tools:** Pandas, NumPy, Git, Jenkins CI/CD, Splunk, Spring Boot

PROFESSIONAL EXPERIENCE

Johns Hopkins University - Office of Digital Transformation — **Baltimore, MD**

Software Engineer Intern — *June 2025 - August 2025*

- Built Python pipelines to validate and standardize multi-source institutional datasets, enhancing data quality for analytics.
- Developed tools for 200+ faculty that reduced manual data processing by 25% and improved the reliability of reporting data.

Barclays — **Pune, India**

Lead Software Engineer — *April 2023 - May 2024*

- Led development of a real-time analytics system processing 1M+ daily transactions, enabling executive-level insights at scale.
- Automated deployment and orchestration pipelines using Autosys and TeamCity, cutting release latency from 4+ hours to under 10 minutes and improving availability of data services.

JPMorgan Chase — **Mumbai, India**

Software Engineer III — *October 2021 - April 2023*

- Developed and scaled a data ingestion and enrichment pipeline using Java and Apache Spark, processing Gigabytes per day of unstructured financial data for cloud-based data lakes and downstream modeling.
- Led migration of terabyte-scale regulated datasets to AWS S3 with integrity checks, ensuring zero data loss and compliance.
- Improved data service resilience through microservice redesign and monitoring, reducing MTTR by 70%.

FIS Global — **Pune, India**

Software Engineer — *July 2017 - October 2021*

- Designed and maintained high-volume data transformation pipelines for middle-office trade processing and analytics.
- Built a static bulk-upload tool, saving 10+ hours per week and reducing data errors by ~90%, earning the Above and Beyond award.

ACADEMIC LEADERSHIP AND TEACHING

- Head Course Assistant, Software System Design (Fall 2025)
- Course Assistant, Operating Systems (Spring 2025)
- ILLD Fellow (Integrative Learning and Life Design, Software Engineer - Python) (Fall 2024 - Present)

EXTRACURRICULAR ACTIVITIES

- Diploma in Bharatanatyam
- Hiking
- Iyengar Yoga & Meditation